

Topic 39: Forces – Different types of forces

Students should:		Guidance <i>iLowerSecondary Curriculum reference</i>
39.1	understand that forces are pushes or pulls that can change the speed of an object or the direction it is moving in, or can change the shape of something	<i>Year 7: Forces – Different types of forces</i>
39.2	understand the difference between contact forces, such as friction, upthrust, air and water resistance, and non-contact forces, such as gravity, magnetism and forces due to static electricity	use examples where the forces occur such as gravity (a non-contact force), which causes parachutists to fall to the ground and air resistance (a contact force) stemming from the parachute, which slows the parachutist down <i>Year 7: Forces – Different types of forces</i>
39.3	know that the unit of force is the newton (N)	<i>Year 7: Forces – Different types of forces</i>
39.4	understand the use of different-sized arrows to indicate the size and direction of action of a force	<i>Year 7: Forces – Different types of forces</i>
39.5	know the difference between mass and weight	appreciate that an object on Earth has weight because of gravity <i>Year 7: Forces – Different types of forces</i>
39.6	understand the causes of friction, air resistance (drag), water resistance (drag) and upthrust	appreciate forces of friction, air resistance and water resistance can be helpful and unhelpful, and how they can be changed understand the term <i>streamlining</i> and how it can be used beneficially to reduce air resistance and water resistance <i>Year 7: Forces – Different types of forces</i>